Chapter 5
Image Restoration

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Preview

- Goal of image restoration
  - Improve an image in some predefined sense
  - Difference with image enhancement ?
- Features
  - Image restoration v.s image enhancement
  - Objective process v.s. subjective process
  - A prior knowledge v.s heuristic process
  - A prior knowledge of the degradation phenomenon is considered
  - Modeling the degradation and apply the inverse process to recover the original image

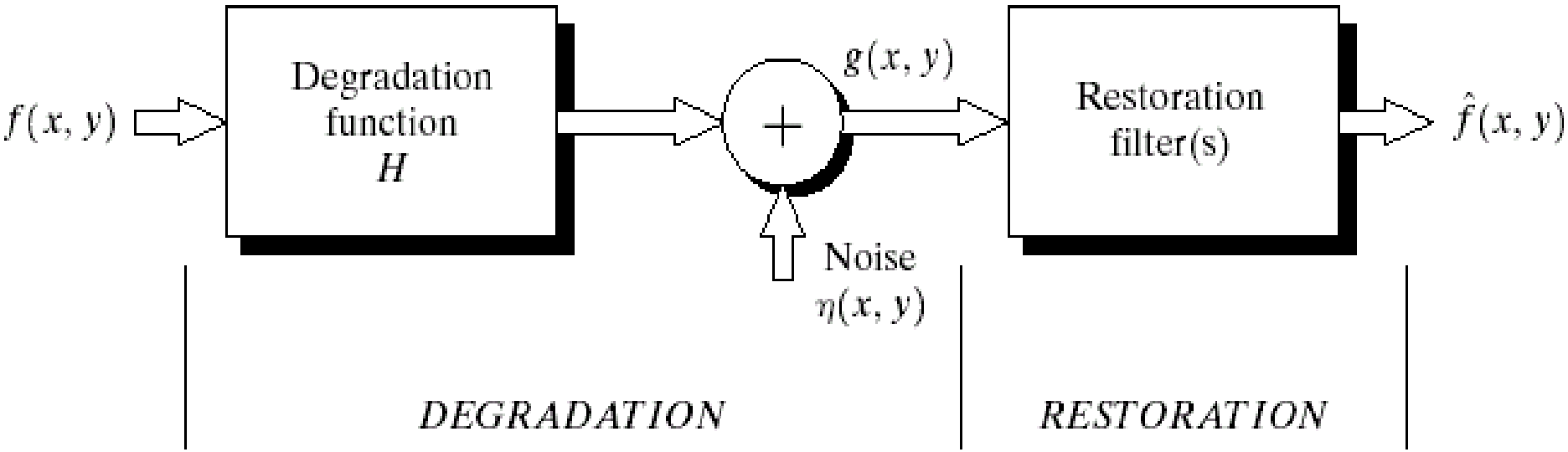
Preview (cont.)

- Target
  - Degraded digital image
  - Sensor, digitizer, display degradations are less considered
- Spatial domain approach
- Frequency domain approach

Outline

- A model of the image degradation / restoration process
- Noise models
- Restoration in the presence of noise only – spatial filtering
- Periodic noise reduction by frequency domain filtering

A model of the image degradation/restoration process



$$\begin{cases} g(x,y)=f(x,y)*h(x,y)+\eta(x,y) \\ G(u,v)=F(u,v)H(u,v)+N(u,v) \end{cases}$$

Noise models

- Source of noise
  - Image acquisition (digitization)
  - Image transmission
- Spatial properties of noise
  - Statistical behavior of the gray-level values of pixels
  - Noise parameters, correlation with the image
- Frequency properties of noise
  - Fourier spectrum
  - Ex. white noise (a constant Fourier spectrum)

Noise probability density functions

- Noises are taken as random variables
- Random variables
  - Probability density function (PDF)

Gaussian noise

- Math. tractability in spatial and frequency domain
- Electronic circuit noise and sensor noise

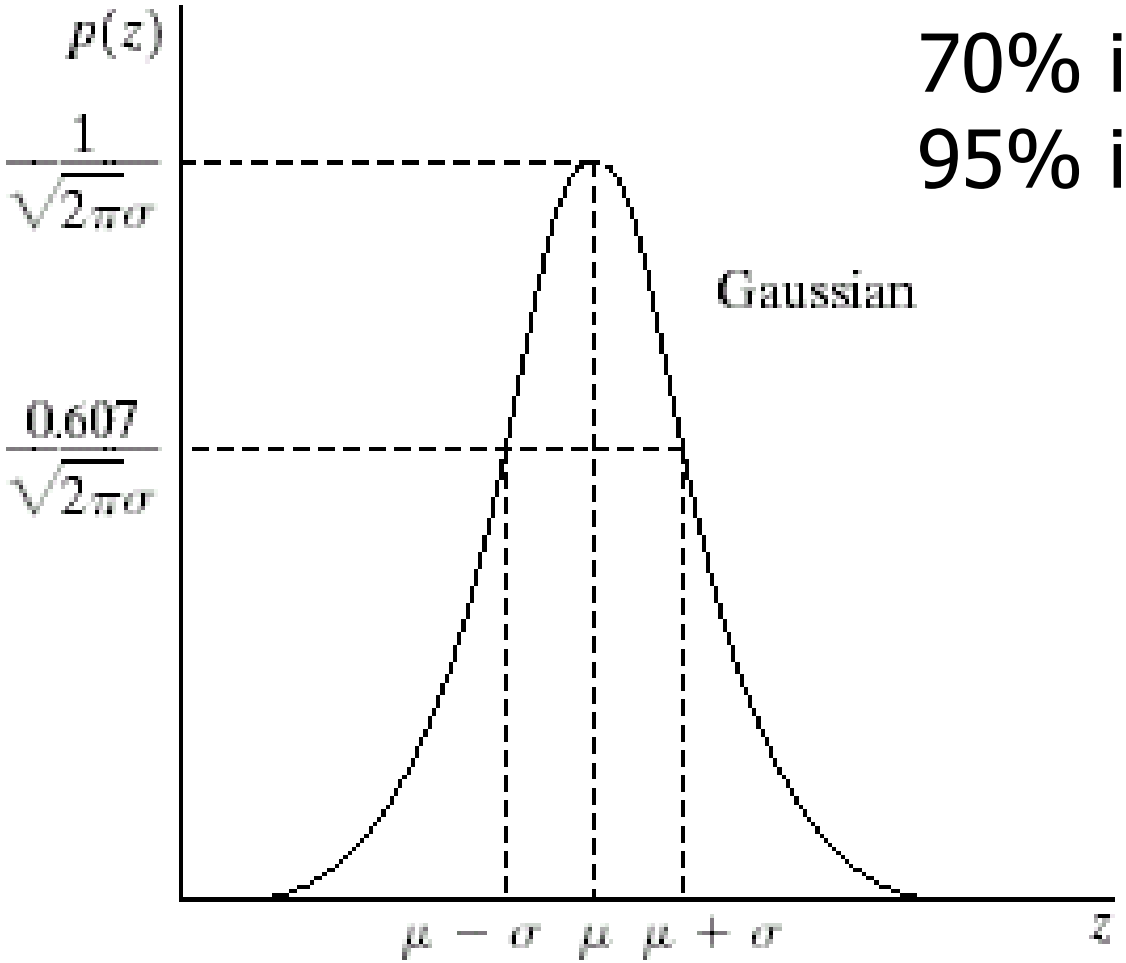
$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\mu)^2/2\sigma^2}$$

mean

variance

Note: $\int_{-\infty}^{\infty} p(z)dz = 1$

Gaussian noise (PDF)



70% in $[(\mu-\sigma), (\mu+\sigma)]$
95% in $[(\mu-2\sigma), (\mu+2\sigma)]$

Uniform noise

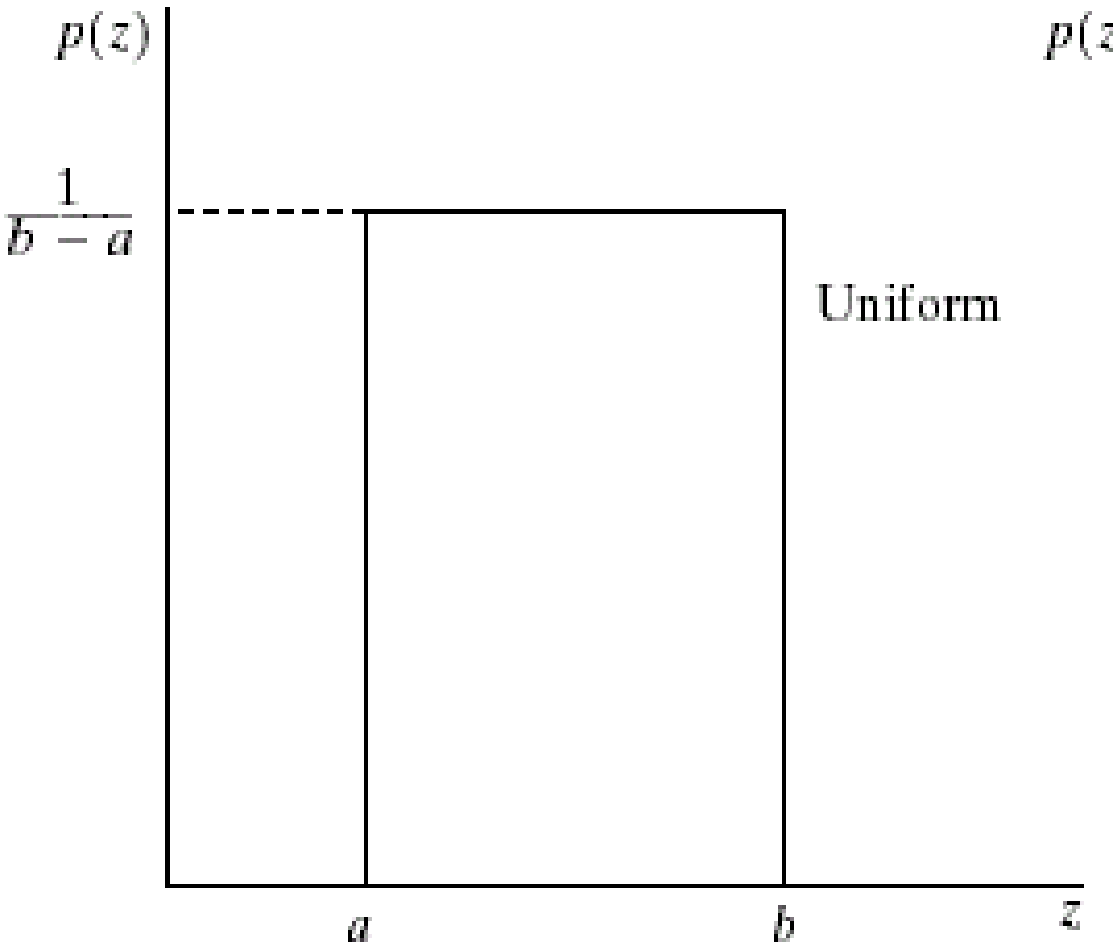
- Less practical, used for random number generator

$$p(z) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le z \le b \\ 0 & \text{otherwise} \end{cases}$$

Mean: $$\mu = \frac{a+b}{2}$$

Variance: $$\sigma^2 = \frac{(b-a)^2}{12}$$

Uniform PDF



Impulse (salt-and-pepper) nosie

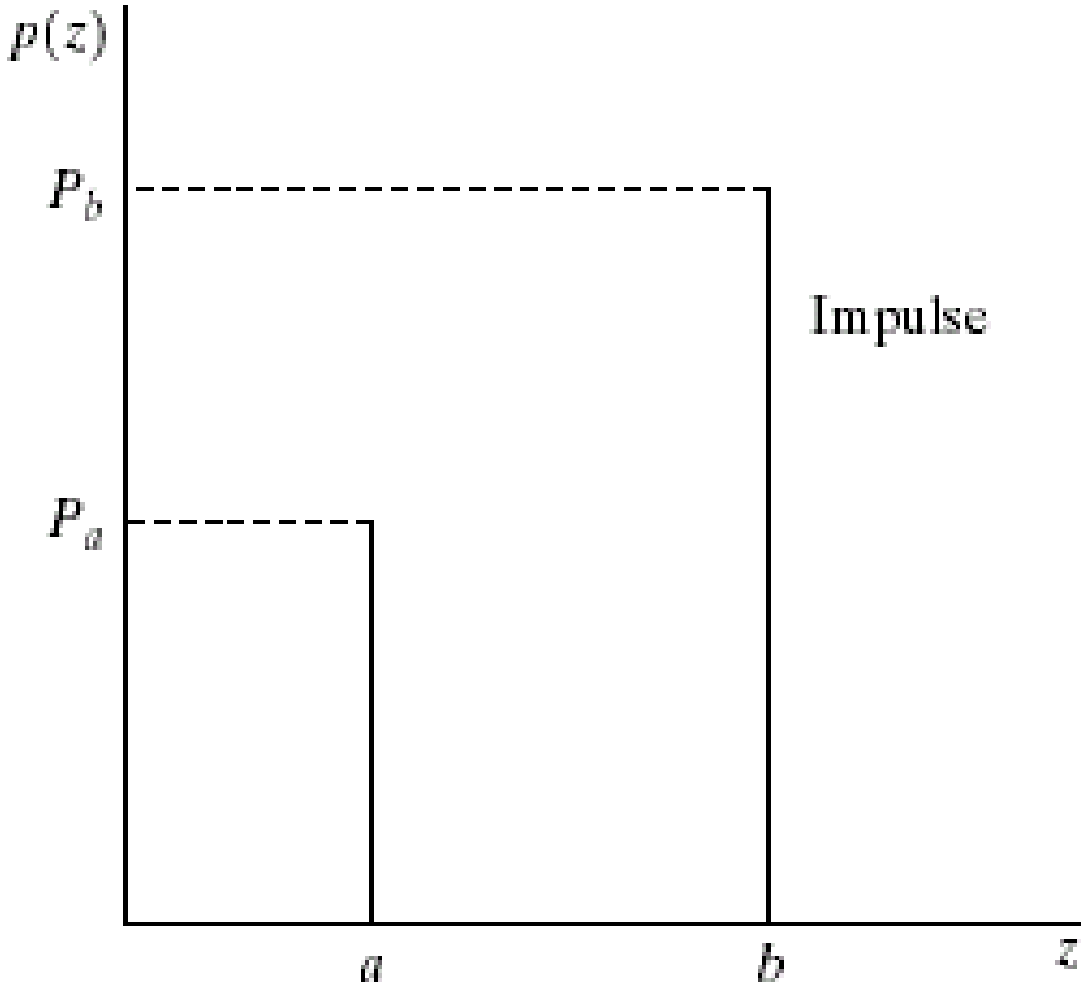
- Quick transients, such as faulty switching during imaging

$$p(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases}$$

If either P_a or P_b is zero, it is called *unipolar.* Otherwise, it is called bipoloar.

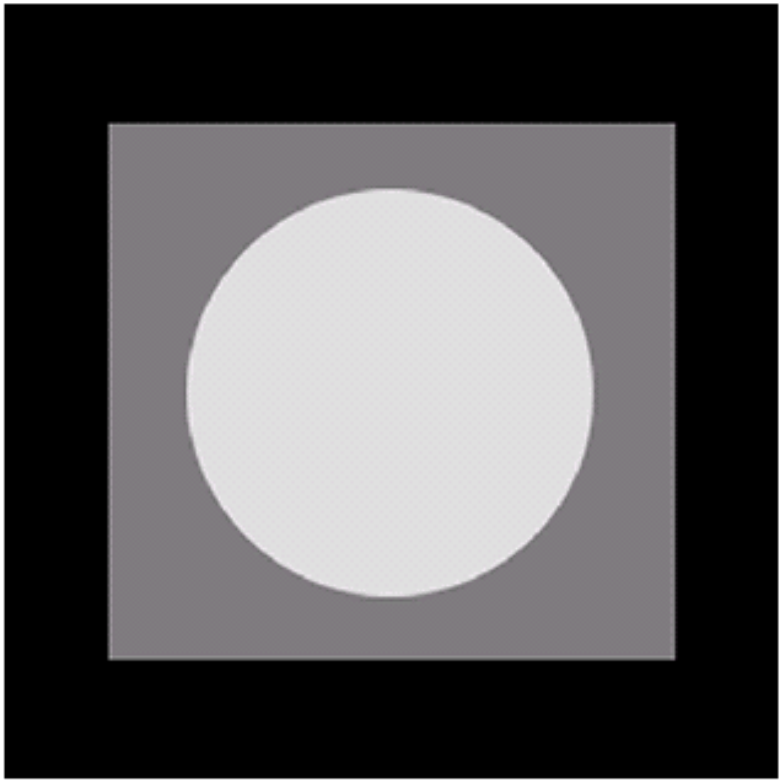
- In practical, impulses are usually stronger than image signals. Ex., a=0(black) and b=255(white) in 8-bit image.

Impulse (salt-and-pepper) nosie PDF

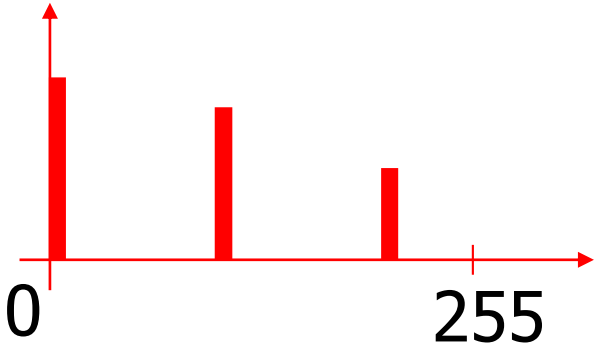


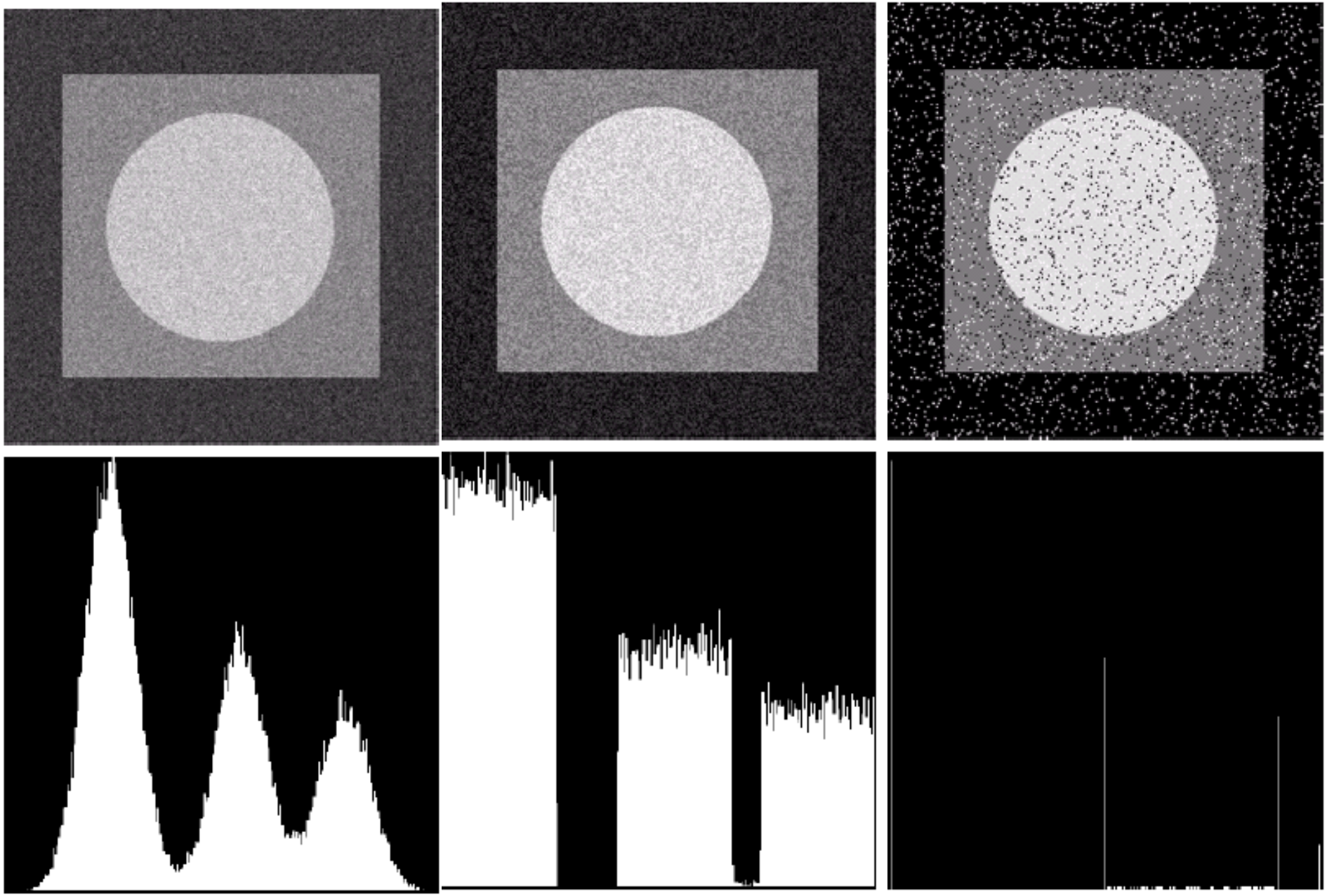
Test for noise behavior

- Test pattern



Its histogram:

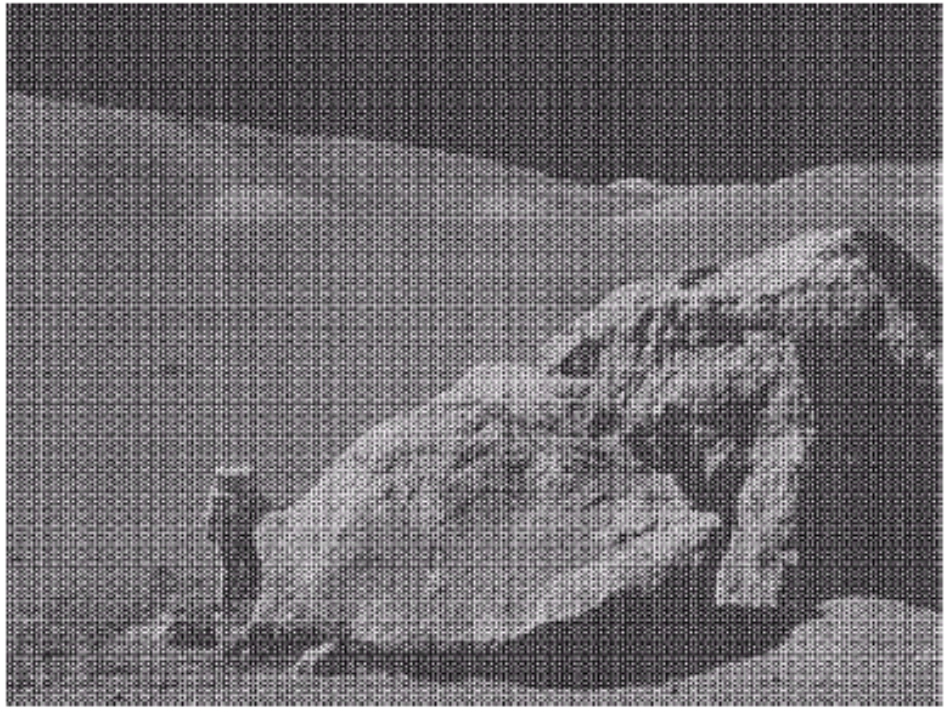


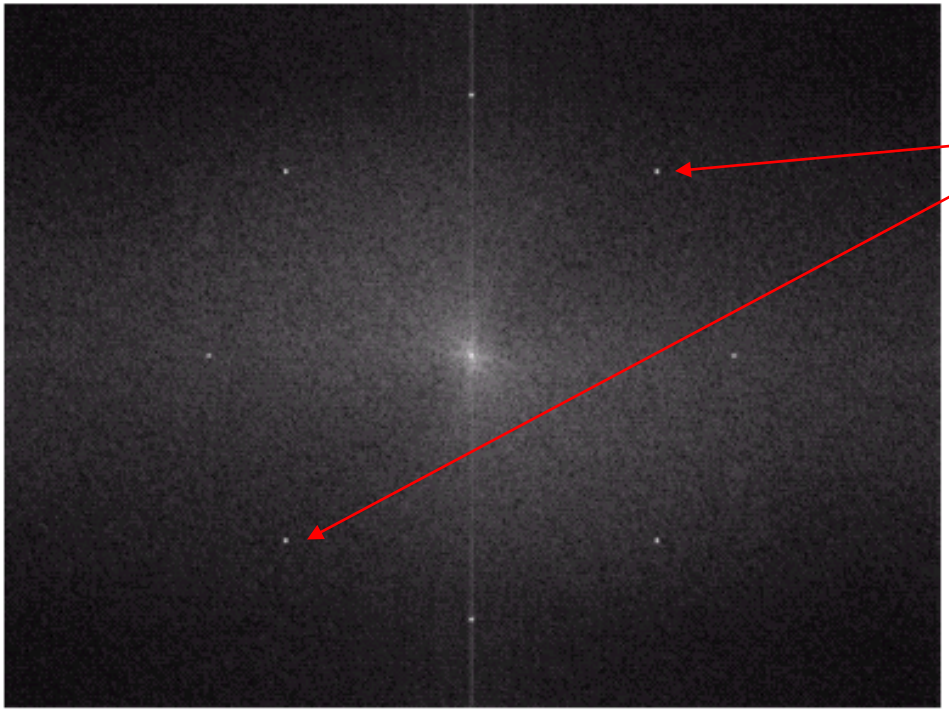


Gaussian Uniform Salt & Pepper

Periodic noise

- Arise from electrical or electromechanical interference during image acquisition
- Spatial dependence
- Observed in the frequency domain



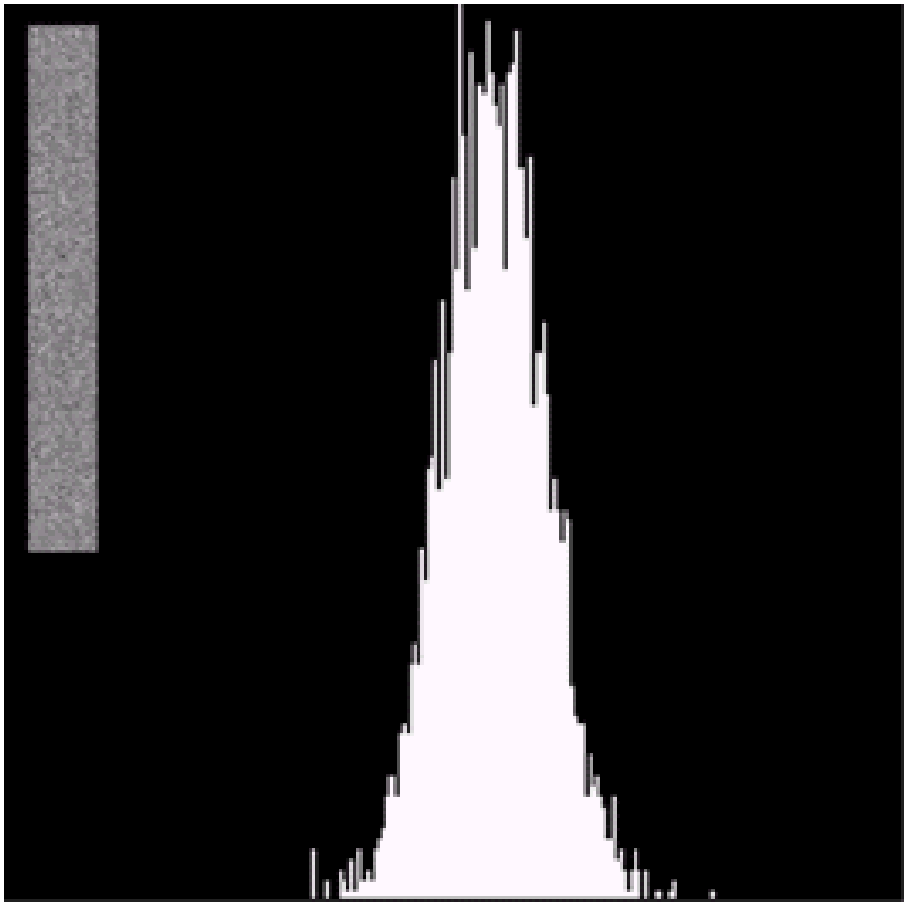


Sinusoidal noise: Complex conjugate pair in frequency domain

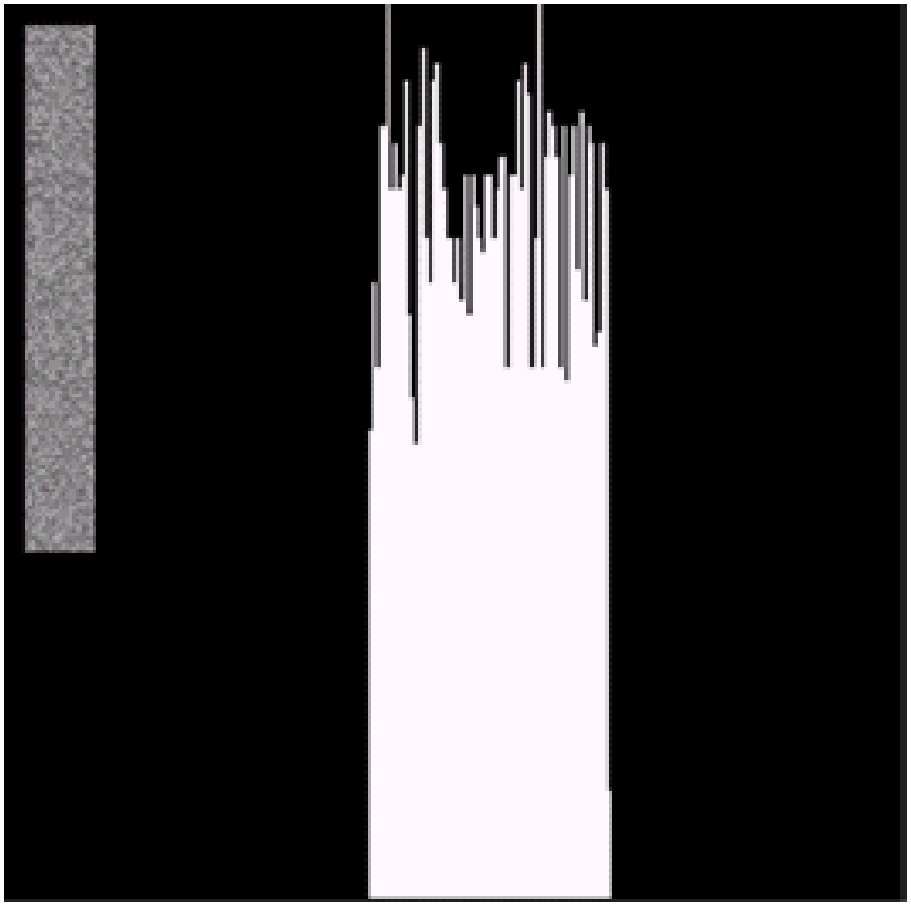
Estimation of noise parameters

- Periodic noise
  - Observe the frequency spectrum
- Random noise with unknown PDFs
  - Case 1: imaging system is available
    - Capture images of "flat" environment
  - Case 2: noisy images available
    - Take a strip from constant area
    - Draw the histogram and observe it
    - Measure the mean and variance

Observe the histogram



Gaussian



uniform

Measure the mean and variance

- Histogram is an estimate of PDF

$$\begin{cases} \mu = \sum_{z_i \in S} z_i \, p(z_i) \\ \sigma^2 = \sum_{z_i \in S} (z_i - \mu)^2 \, p(z_i) \end{cases}$$

$\Leftrightarrow$ $\begin{cases} \text{Gaussian: } \mu, \sigma \\ \text{Uniform: } a, b \end{cases}$

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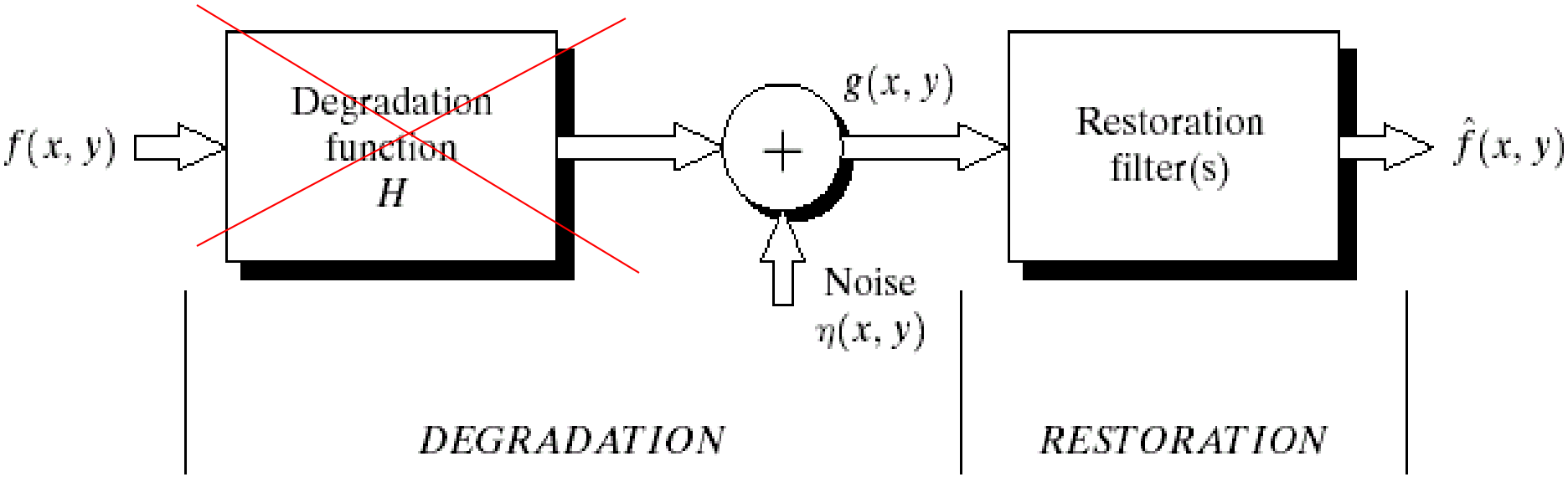
- Linear, position-invariant degradations
- Estimating the degradation function
- Inverse filtering

Additive noise only



$$\begin{cases} g(x,y)=f(x,y)+\eta(\mathbf{x,y}) \\ G(u,v)=F(u,v)+N(u,v) \end{cases}$$

Spatial filters for de-noising additive noise

- Skills similar to image enhancement
- Mean filters
- Order-statistics filters
- Adaptive filters

Mean filters

- Arithmetic mean

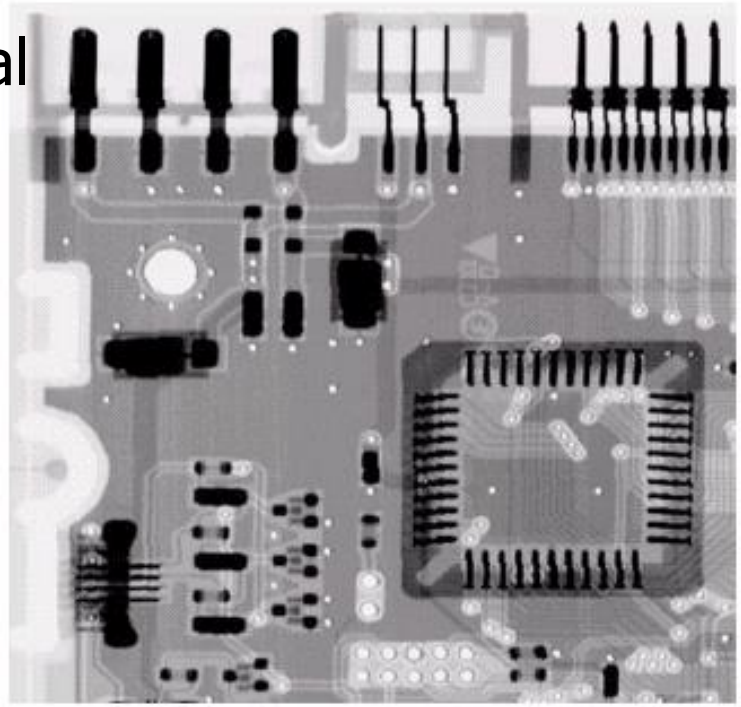
$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t)\in S_{xy}} g(s,t)$$

Window centered at (x,y)

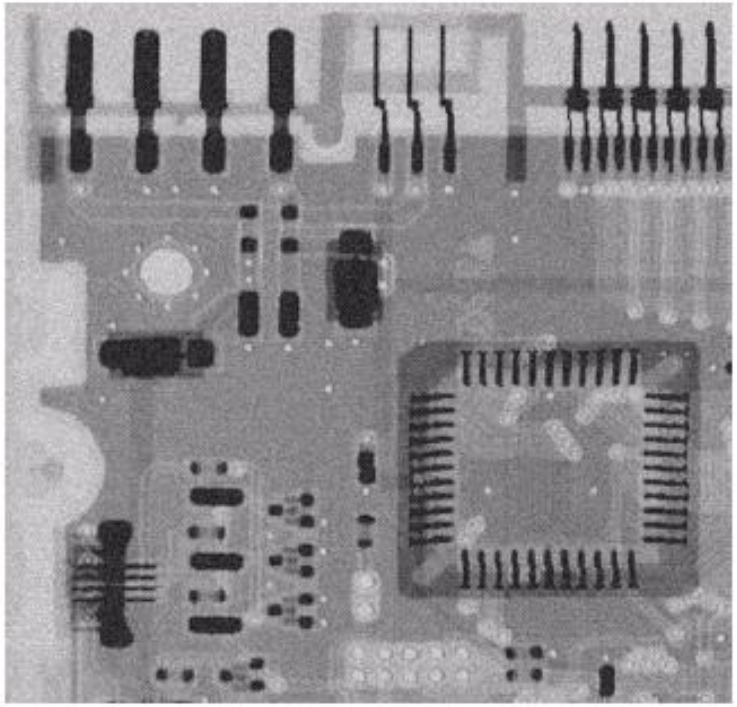
- Geometric mean

$$\hat{f}(x,y) = \left[\prod_{(s,t)\in S_{xy}} g(s,t) \right]^{1/mn}$$

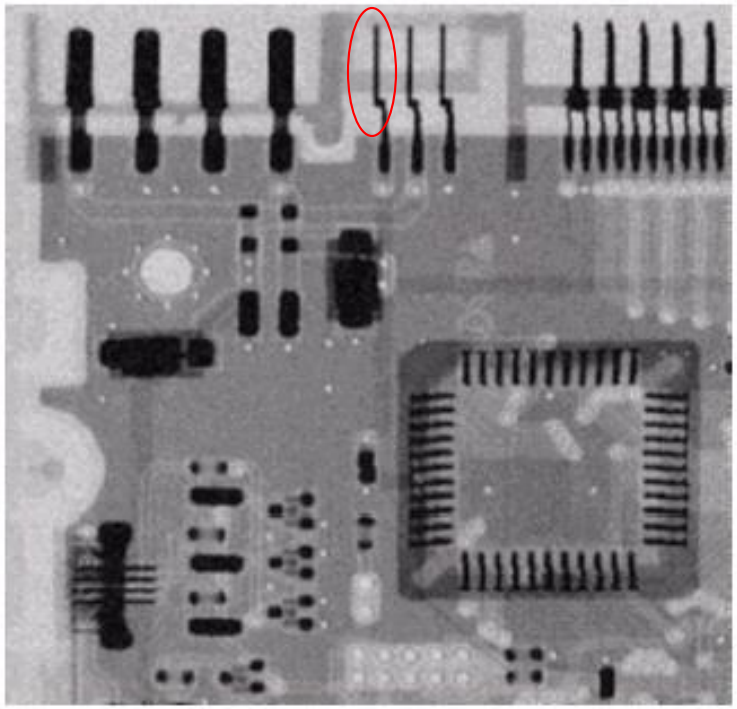
original



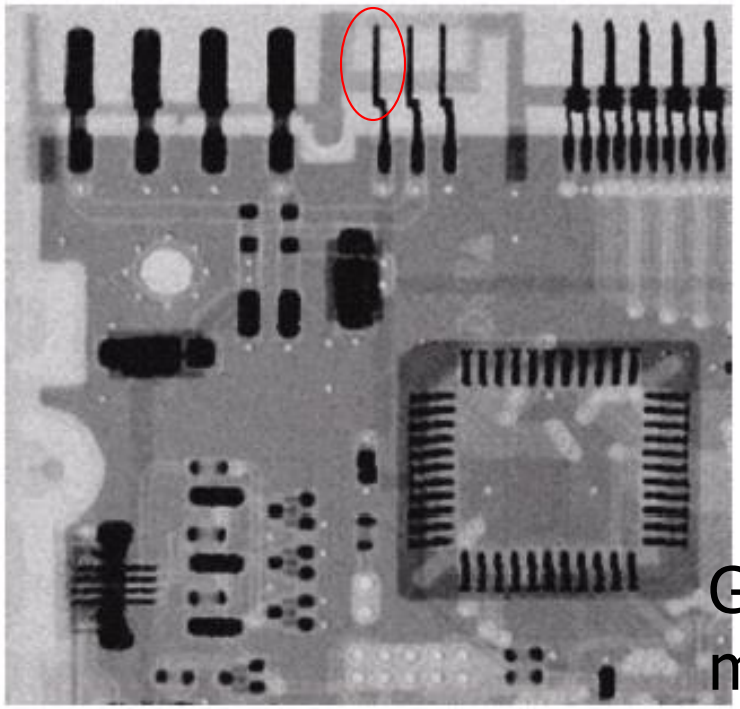
Noisy Gaussian
$\mu=0$
$\sigma=20$



Arith. mean



Geometric mean



Mean filters (cont.)

- Harmonic mean filter

$$\hat{f}(x,y) = \frac{mn}{\sum_{(s,t)\in S_{xy}} \frac{1}{g(s,t)}}$$

- Contra-harmonic mean filter

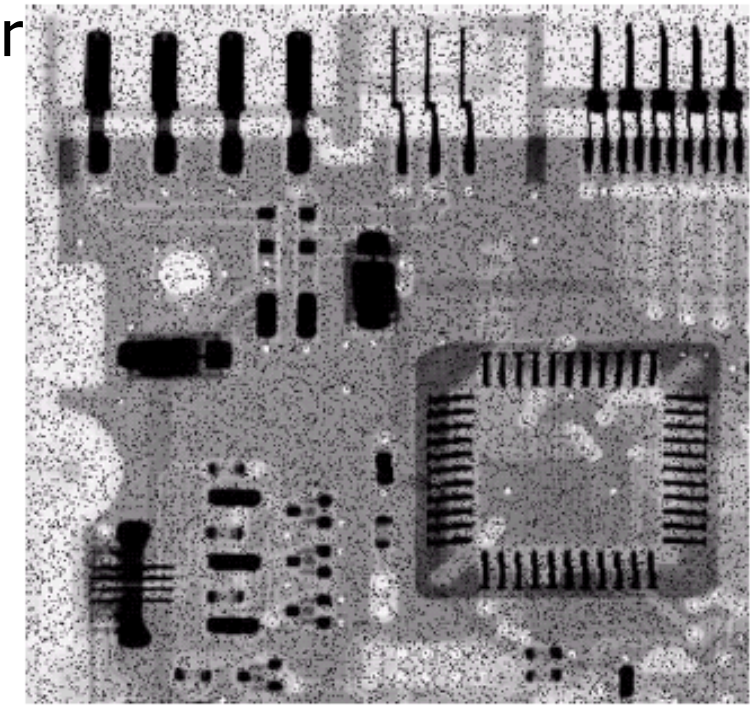
$$\hat{f}(x,y) = \frac{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q}}$$

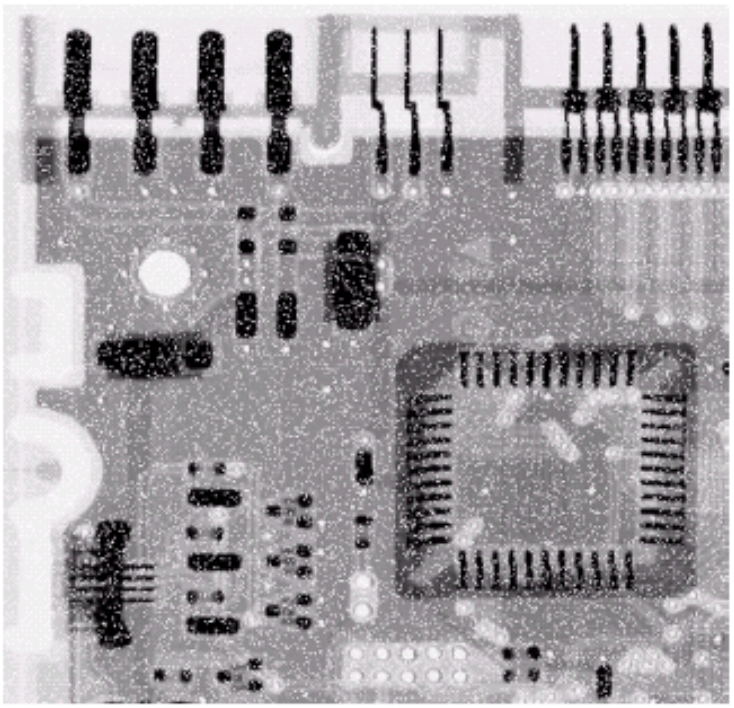
Q=-1, harmonic, Salt Noise

Q=0, Arith. mean

Q=+, Pepper Noise

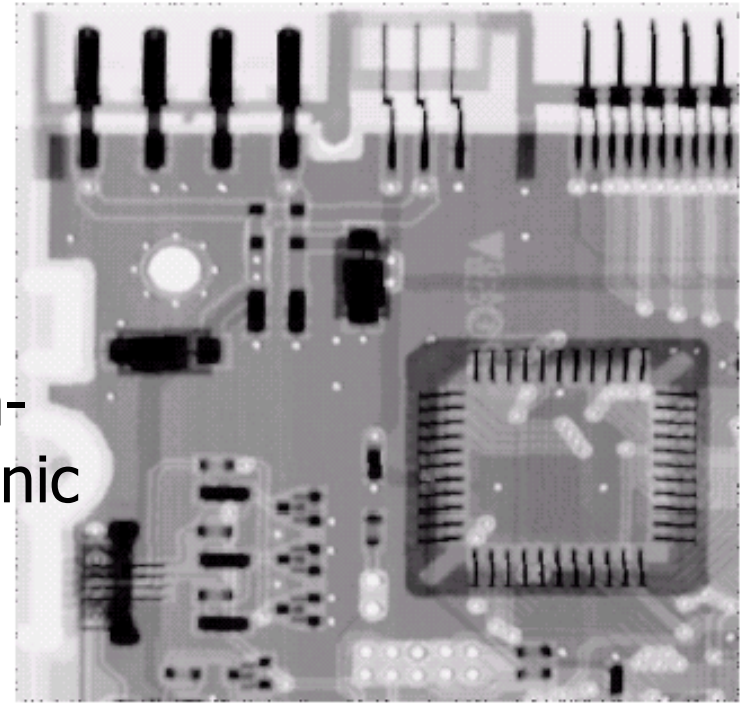
Pepper Noise

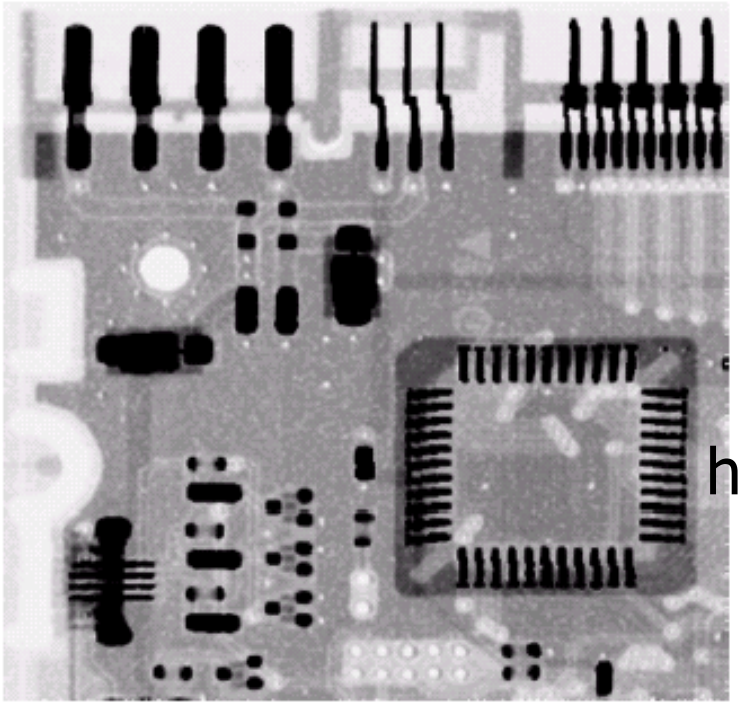




Salt Noise

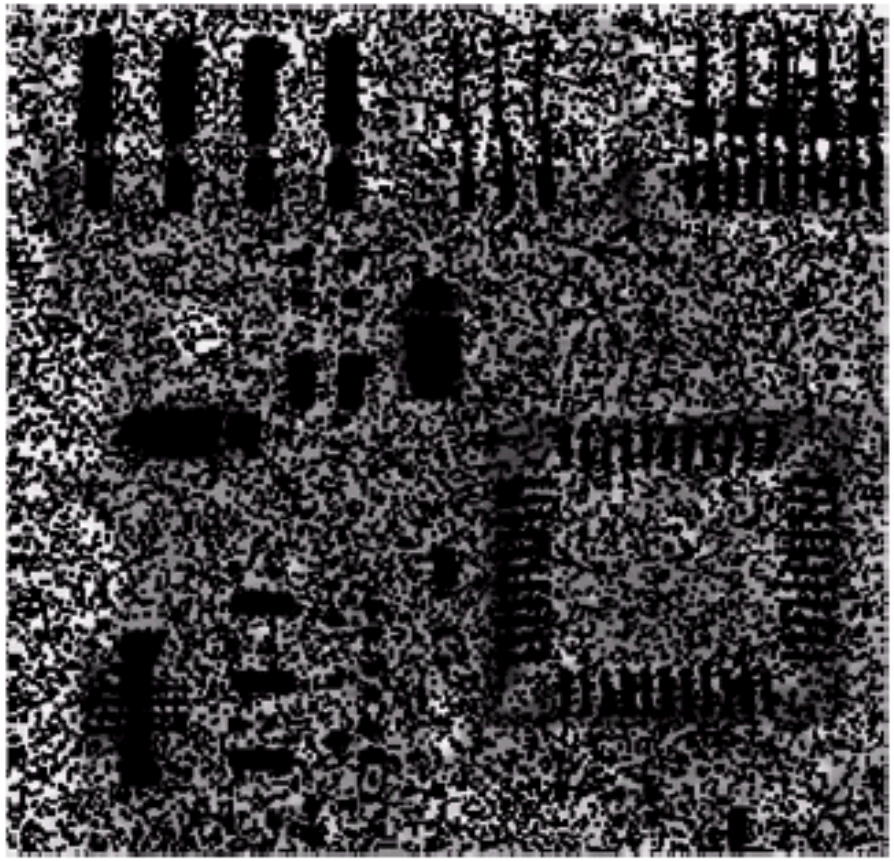
Contra-harmonic Q=1.5



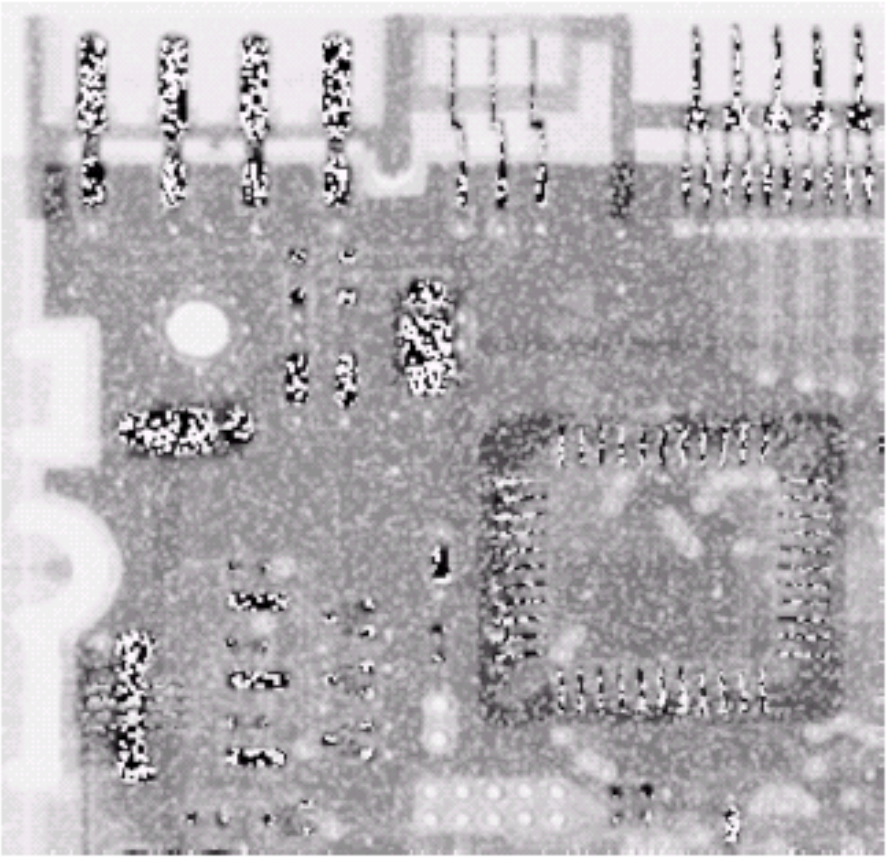


Contra-harmonic Q=-1.5

Wrong sign in contra-harmonic filtering



Q=-1.5



Q=1.5

Order-statistics filters

- Based on the ordering(ranking) of pixels
  - Suitable for unipolar or bipolar noise (salt and pepper noise)
- Median filters
- Max/min filters
- Midpoint filters
- Alpha-trimmed mean filters

Order-statistics filters

- Median filter

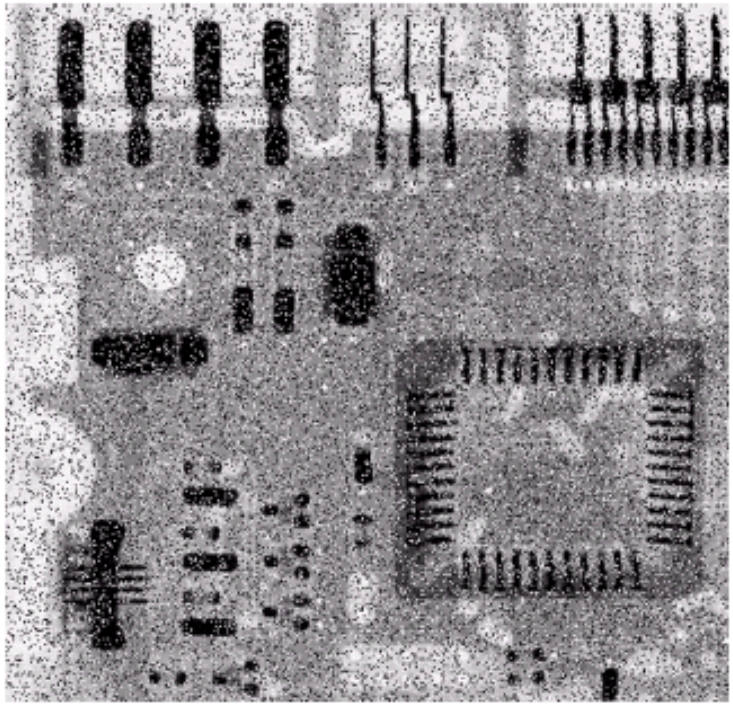
$$\hat{f}(x,y) = \underset{(s,t)\in S_{xy}}{median}\{g(s,t)\}$$

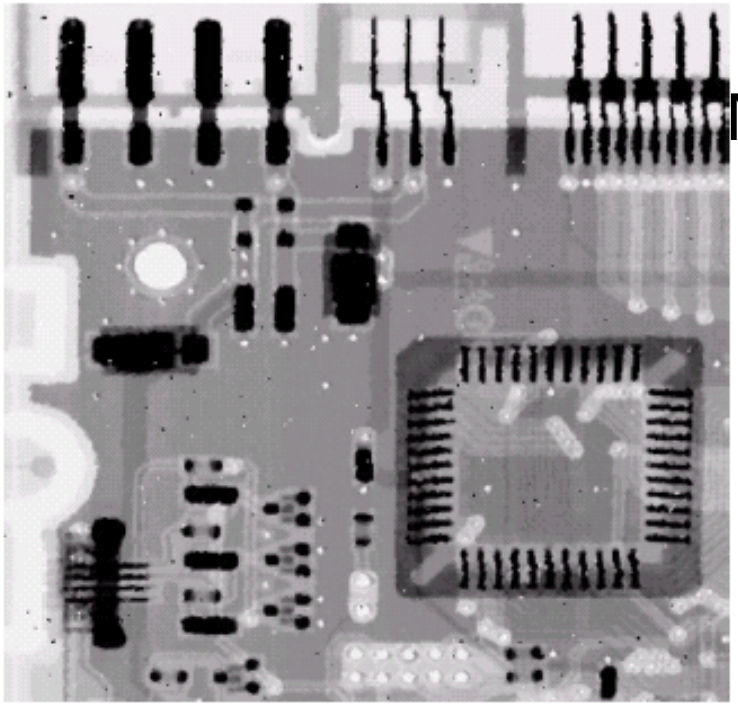
- Max/min filters

$$\hat{f}(x,y) = \underset{(s,t)\in S_{xy}}{\max}\{g(s,t)\}$$

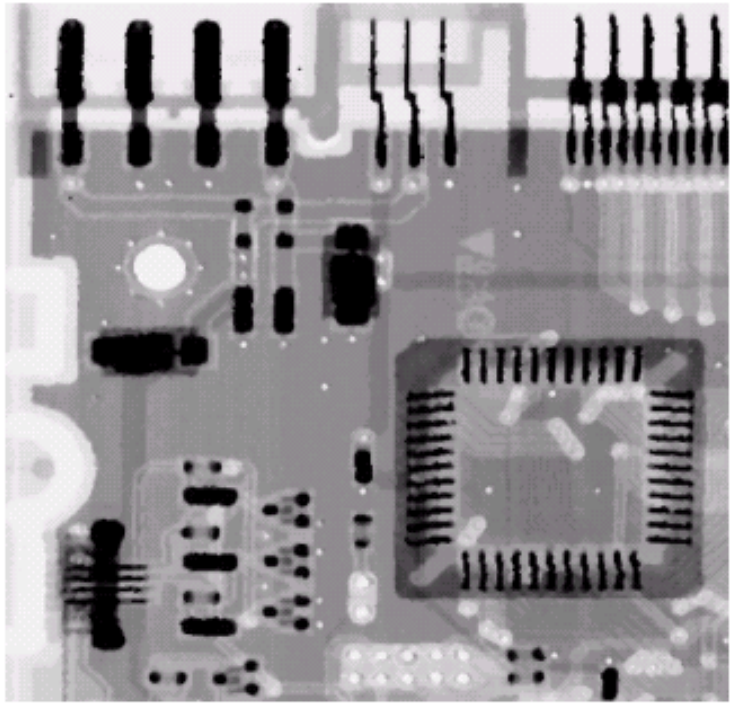
$$\hat{f}(x,y) = \underset{(s,t)\in S_{xy}}{\min}\{g(s,t)\}$$

bipolar Noise
$P_a = 0.1$
$P_b = 0.1$

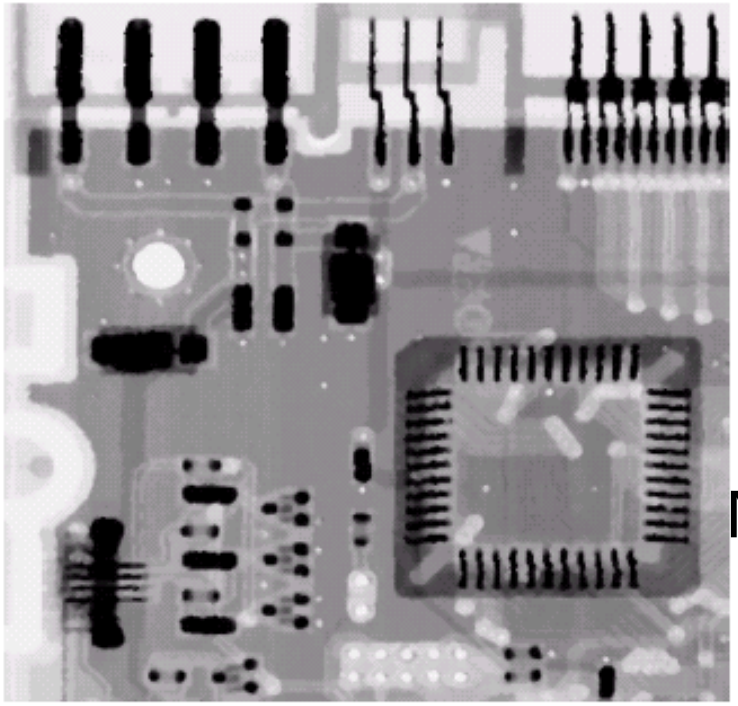




3x3 Median Filter Pass 1

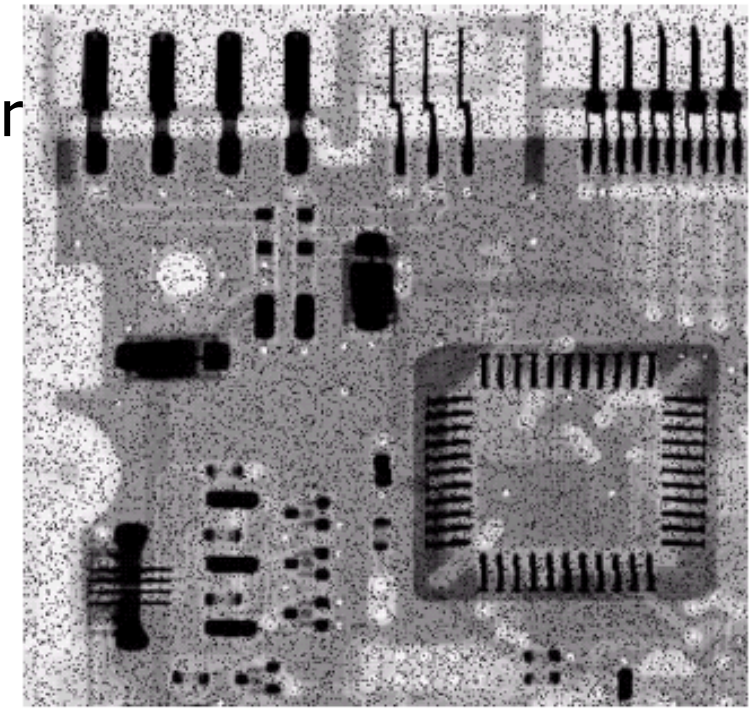


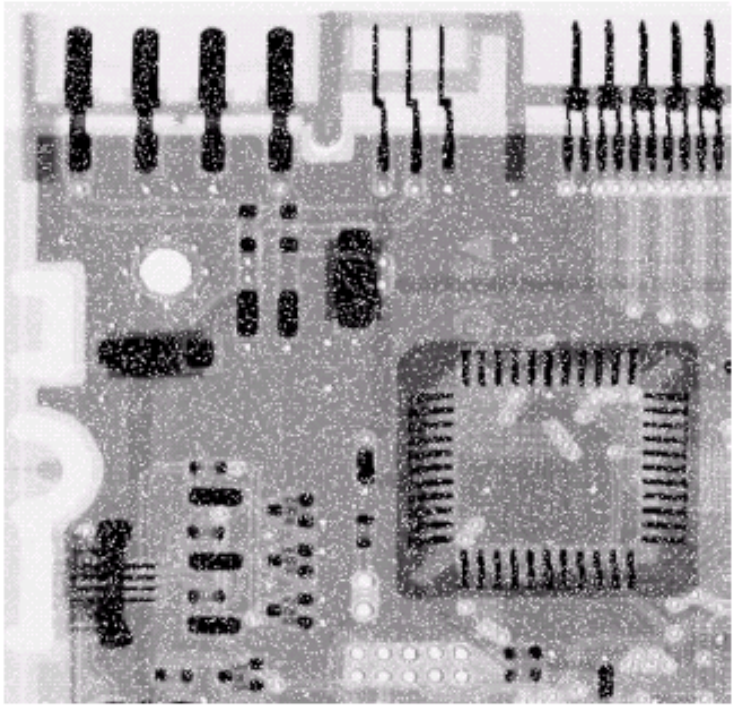
3x3 Median Filter Pass 2



3x3 Median Filter Pass 3

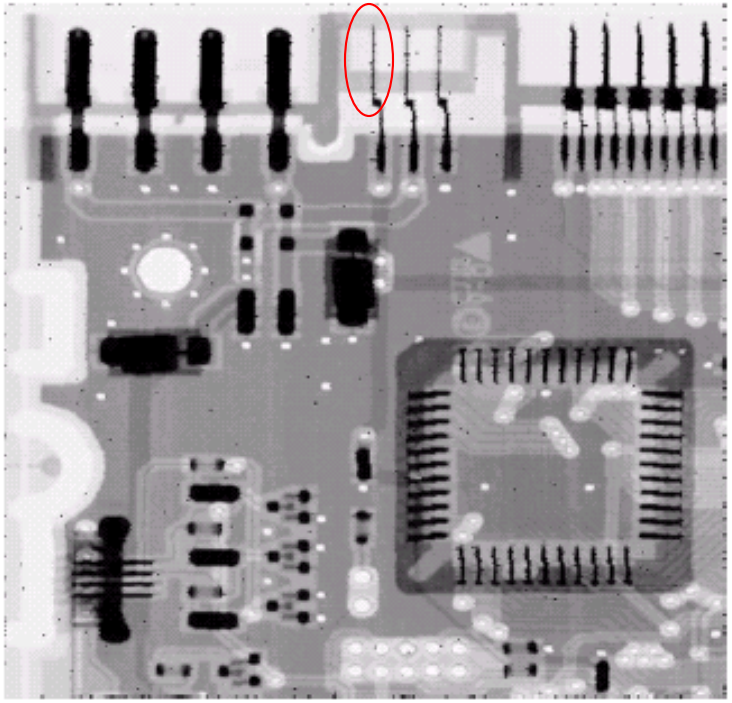
Pepper noise

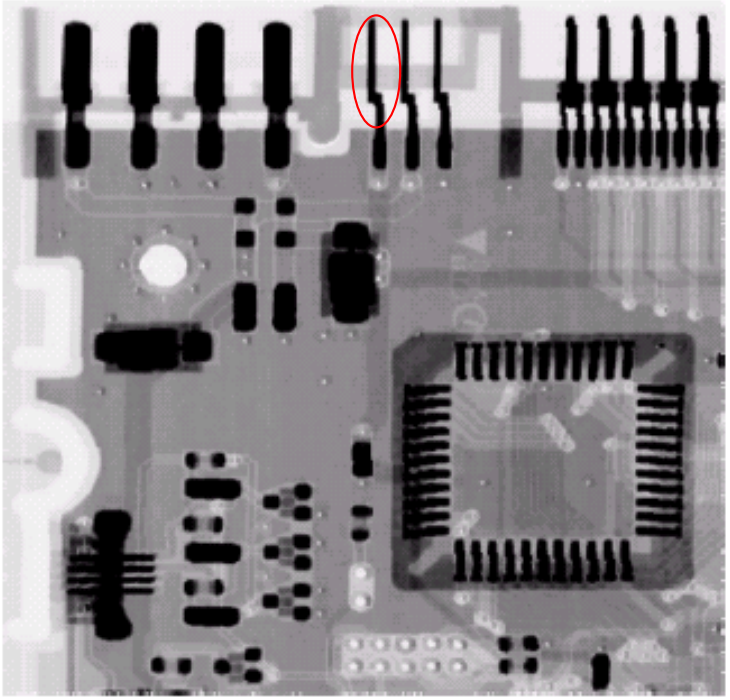




Salt noise

Max filter





Min filter

Order-statistics filters (cont.)

- Midpoint filter

$$\hat{f}(x,y) = \frac{1}{2}\left[\max_{(s,t)\in S_{xy}}\{g(s,t)\} + \min_{(s,t)\in S_{xy}}\{g(s,t)\}\right]$$

- Alpha-trimmed mean filter
  - Delete the d/2 lowest and d/2 highest gray-level pixels

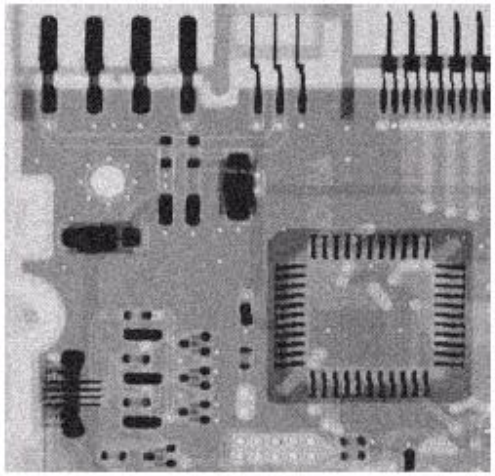
$$\hat{f}(x,y) = \frac{1}{mn-d}\sum_{(s,t)\in S_{xy}} g_r(s,t)$$

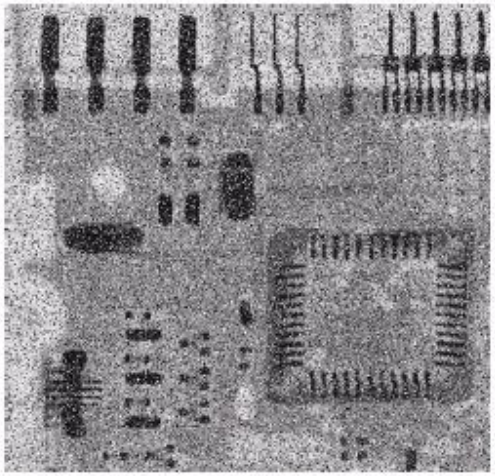
Middle (mn-d) pixels

Uniform noise

$\mu=0$

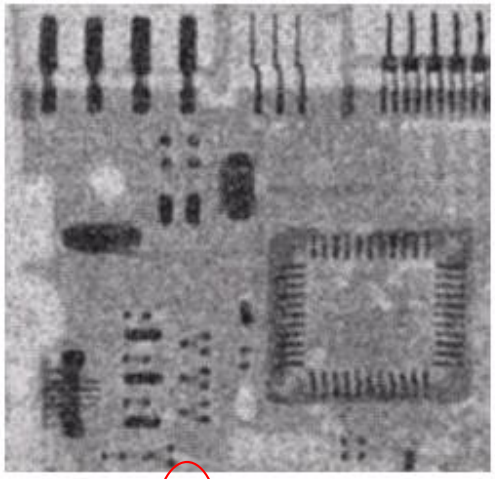
$\sigma^2=800$

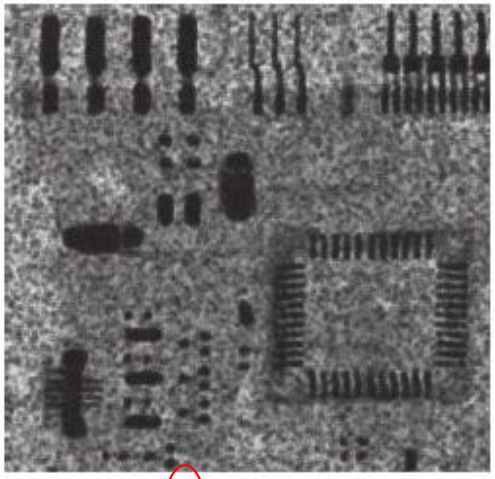




Left +
Bipolar Noise
$P_a = 0.1$
$P_b = 0.1$

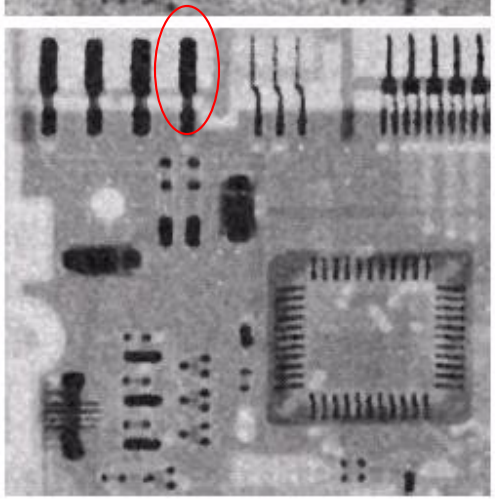
5x5
Arith. Mean
filter

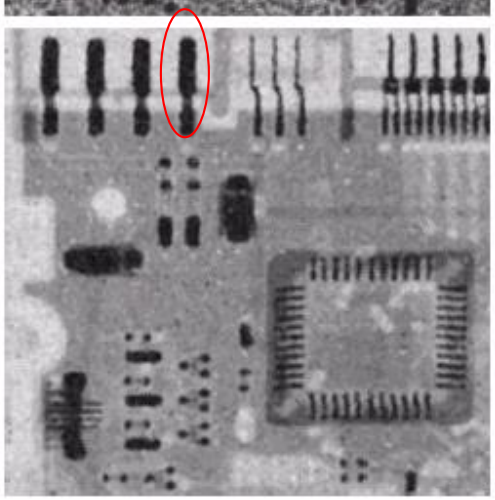




5x5
Geometric
mean

5x5
Median
filter





5x5
Alpha-trim.
Filter
d=5

Outline

- A model of the image degradation / restoration process
- Noise models
- Restoration in the presence of noise only – spatial filtering
- Periodic noise reduction by frequency domain filtering
- Linear, position-invariant degradations
- Estimating the degradation function
- Inverse filtering

Periodic noise reduction

- Pure sine wave
  - Appear as a pair of impulse (conjugate) in the frequency domain

$$f(x,y) = A\sin(u_0 x + v_0 y)$$

$$F(u,v) = -j\frac{A}{2}\left[\delta(u - \frac{u_0}{2\pi}, v - \frac{v_0}{2\pi}) - \delta(u + \frac{u_0}{2\pi}, v + \frac{v_0}{2\pi})\right]$$

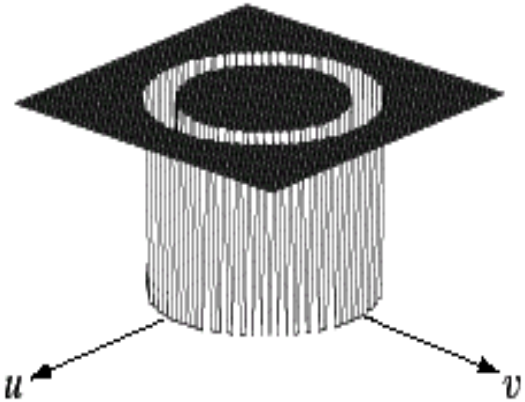
Periodic noise reduction (cont.)

- Bandreject filters
- Bandpass filters
- Notch filters
- Optimum notch filtering

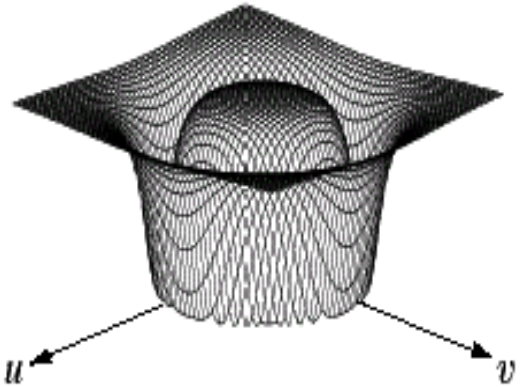
Bandreject filters

* Reject an isotropic frequency

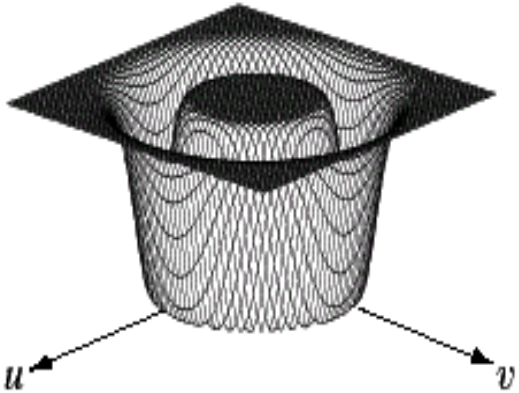
ideal



Butterworth

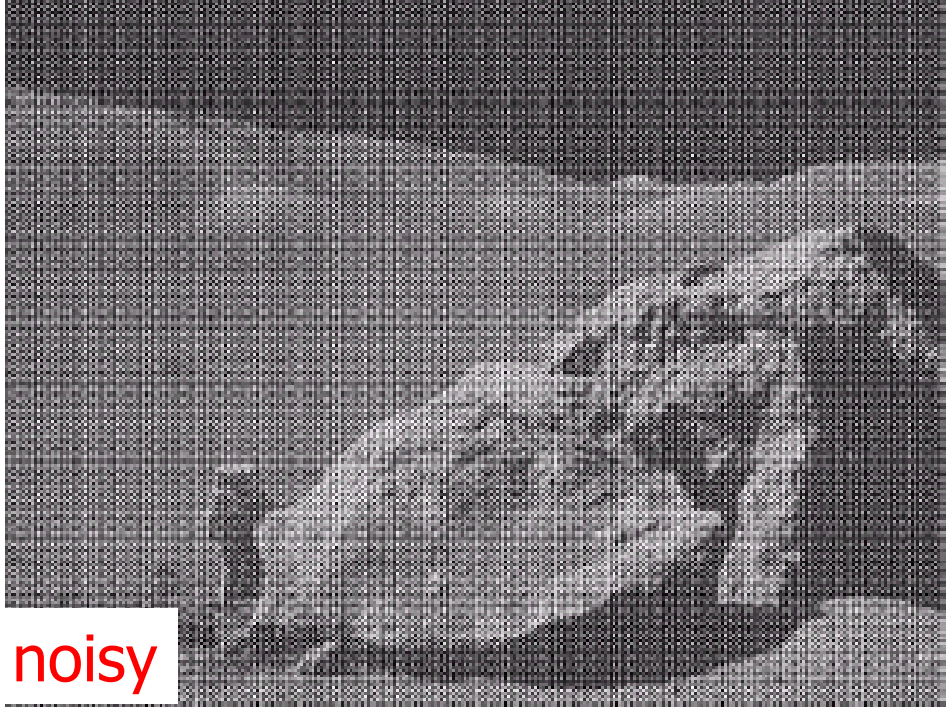


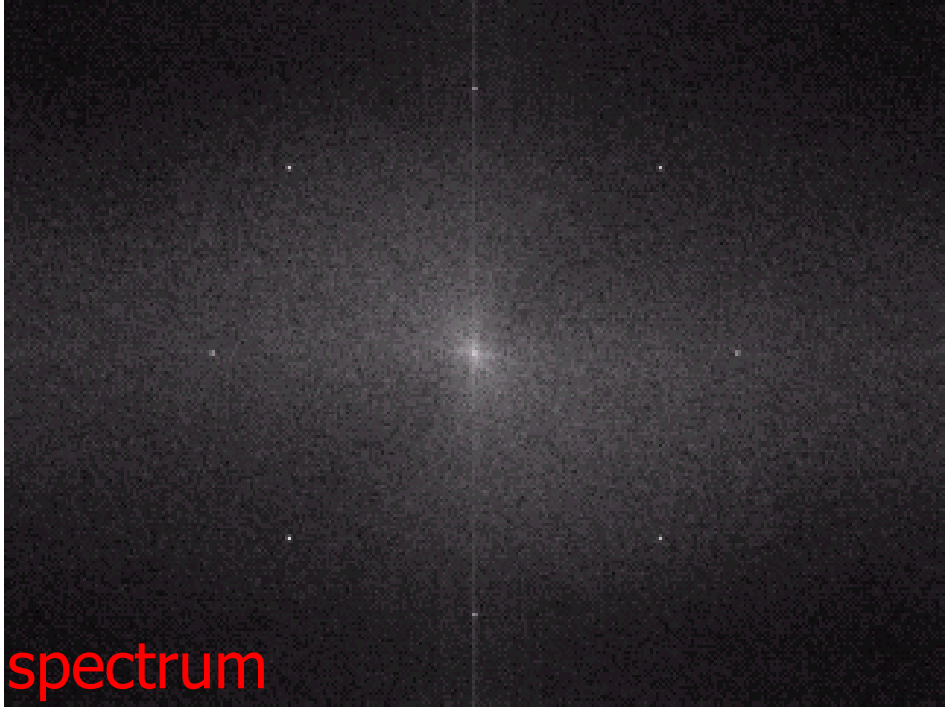
Gaussian

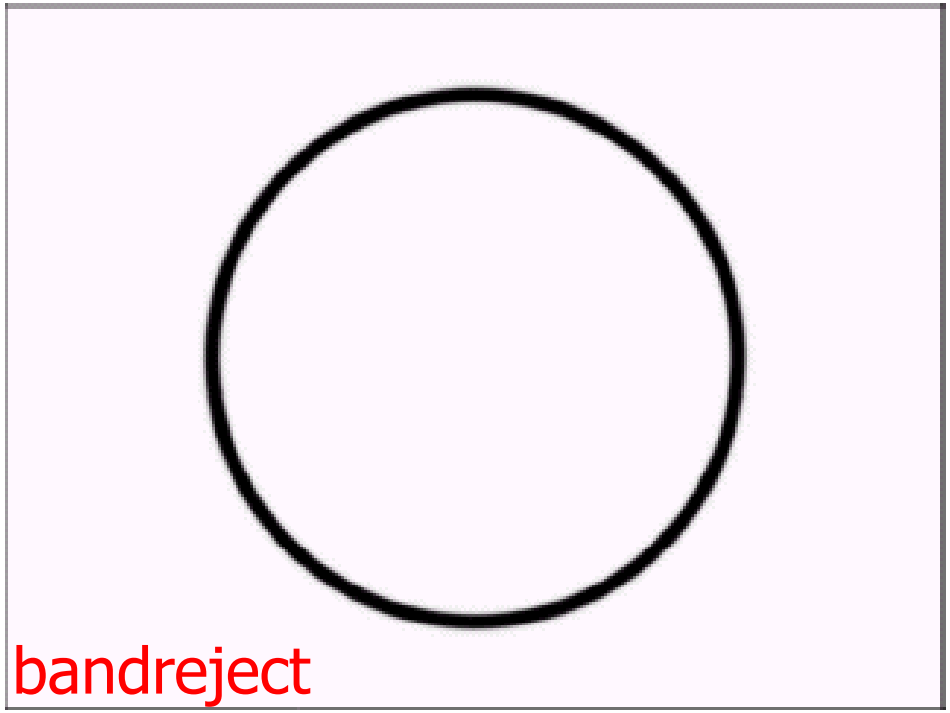


a b c

FIGURE 5.15 From left to right, perspective plots of ideal, Butterworth (of order 1), and Gaussian bandreject filters.


noisy

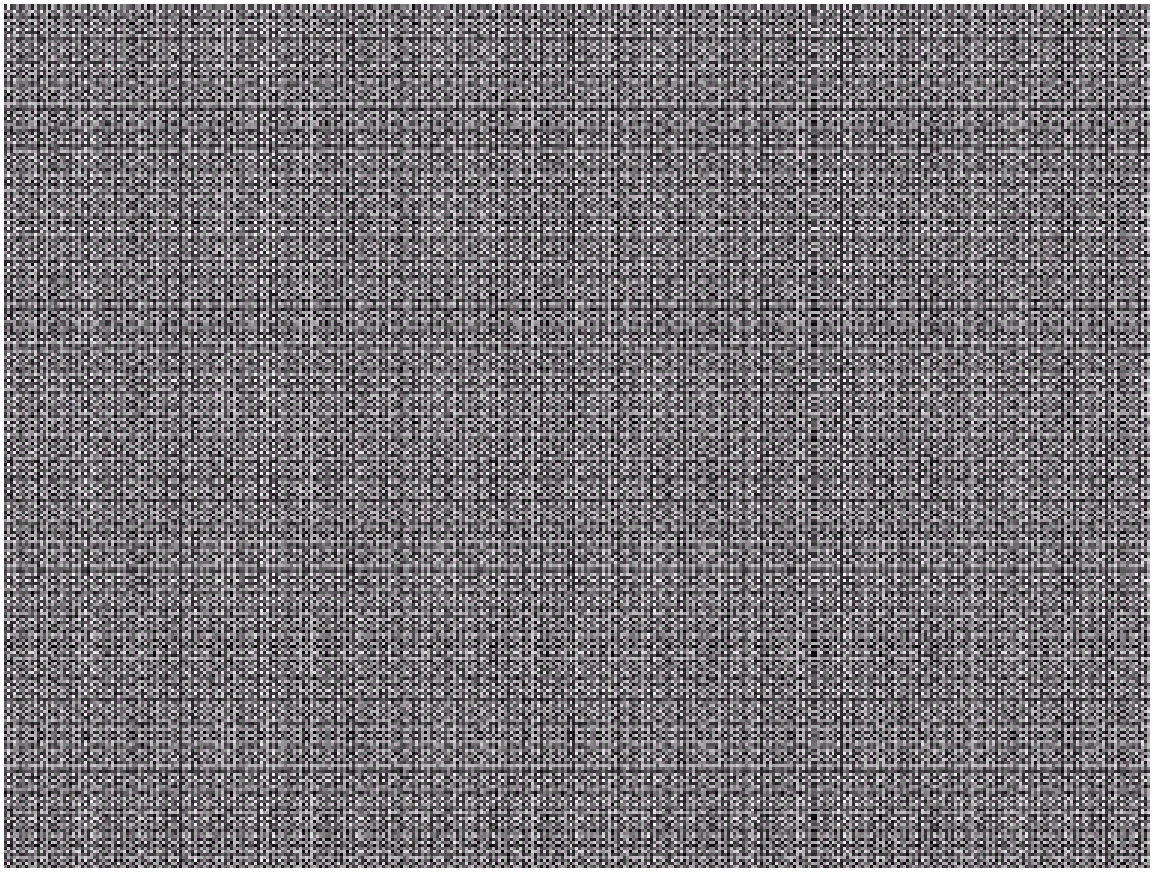

spectrum


bandreject

filtered

Bandpass filters

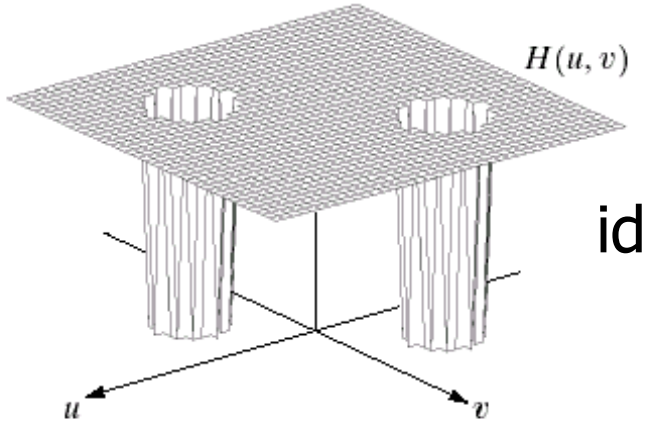
- $H_{bp}(u,v)=1- H_{br}(u,v)$



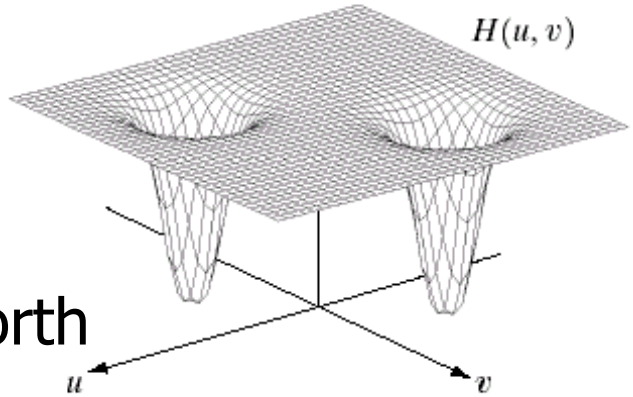
$$\Im^{-1}\{G(u,v)H_{bp}(u,v)\}$$

Notch filters

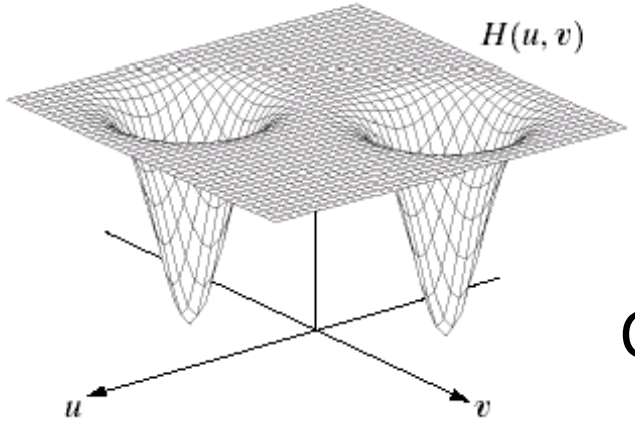
- Reject(or pass) frequencies in predefined neighborhoods about a center frequency

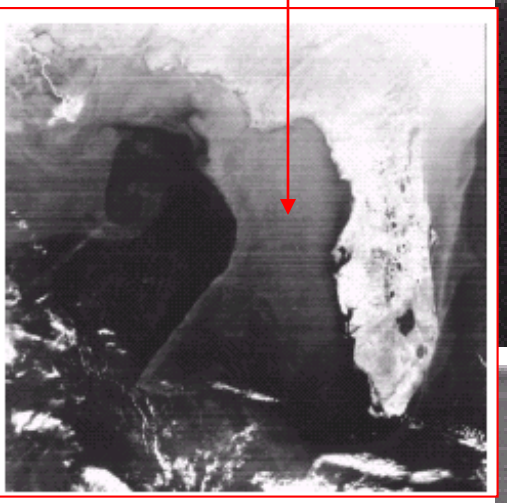
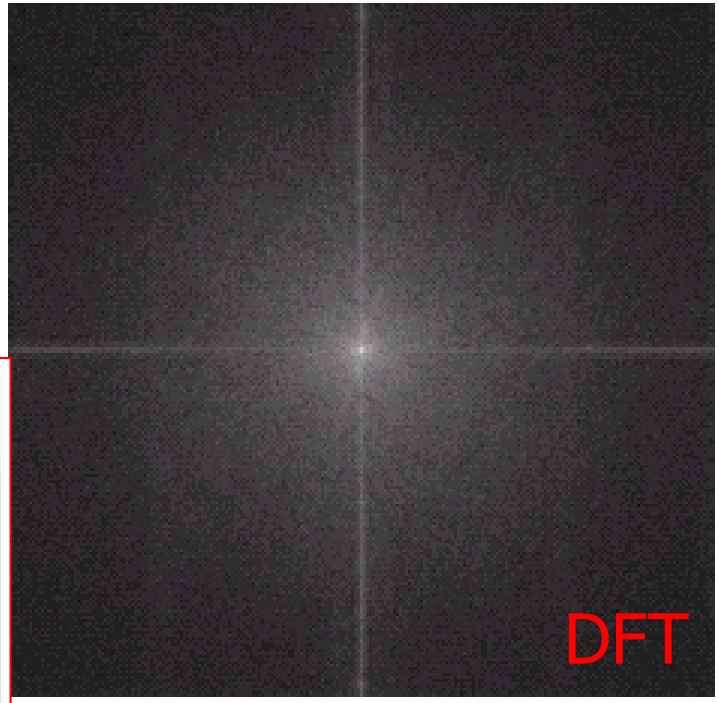
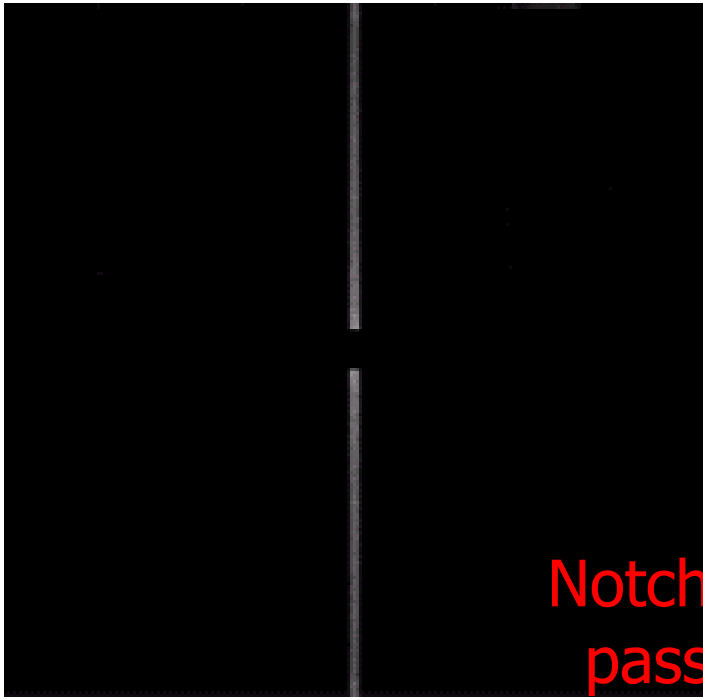
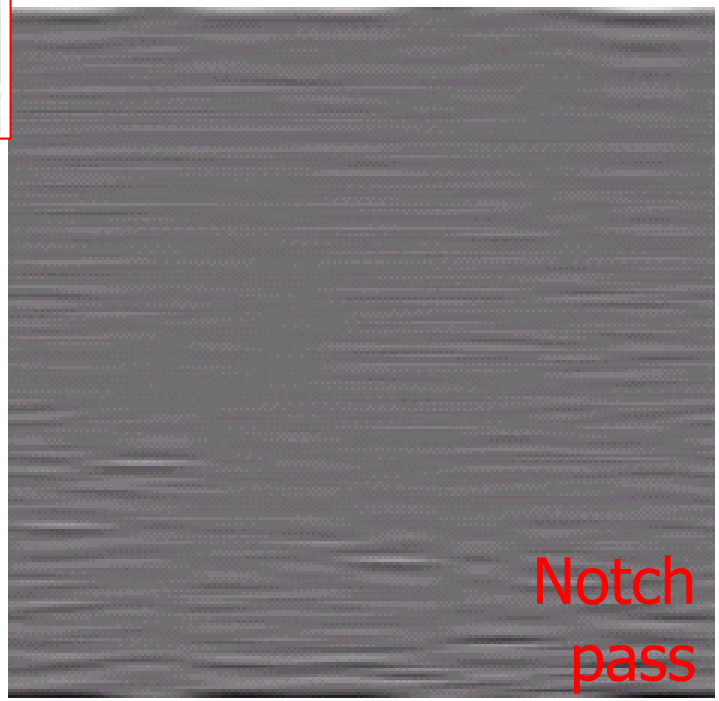
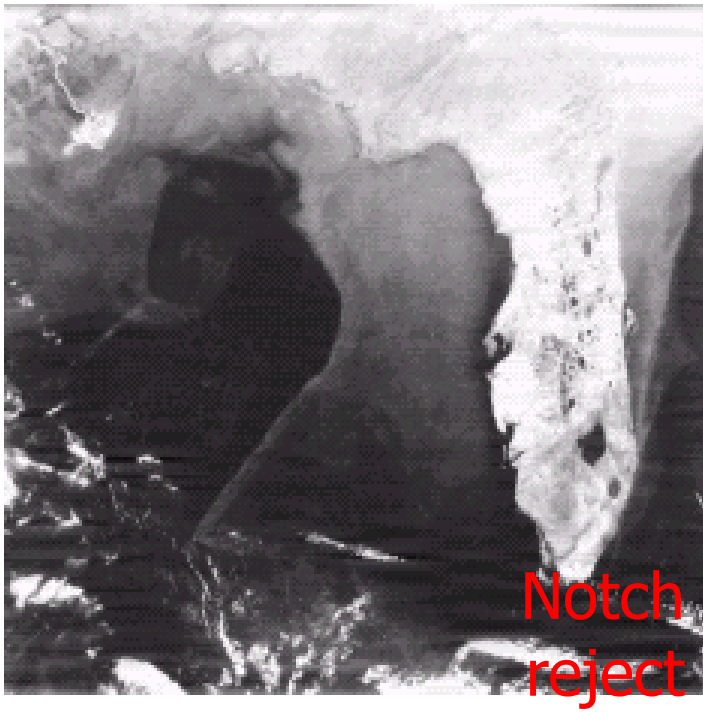


ideal



Butterworth



Gaussian

Horizontal
Scan lines